

A close-up photograph of a large number of ants, likely red and black, moving across a light-colored, sandy or silty surface. The ants are concentrated in a diagonal line from the top right towards the bottom left, with many smaller groups scattered around. The text 'Ant Colony Optimazation Algrothims' is overlaid in a bold, blue, sans-serif font, centered horizontally and slightly lower vertically.

Ant Colony Optimazation Algrothims

Ant stupid, how ant find shortest path to food?



Ant fun facts

- Blind ah hell
- Stupid a hell
- Can spray pheromones/ which “vaporize” with time



Real world applications

Why would ACO's be useful? Why not just use a traditional search algorithm?

- Extremely adaptive, work well in changing environments
- Work well with really complex pathfinding.

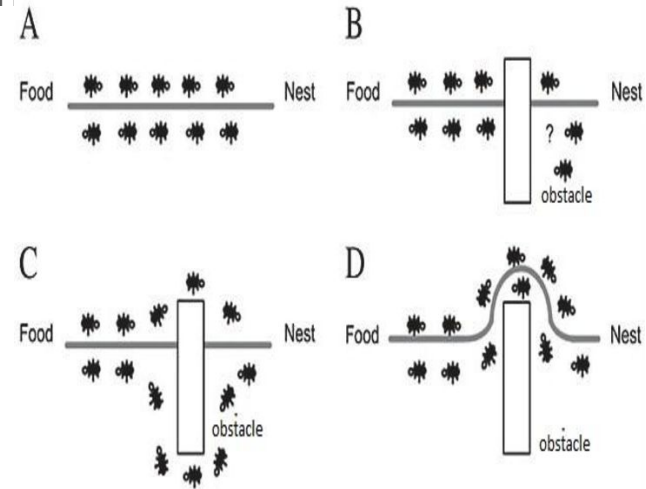
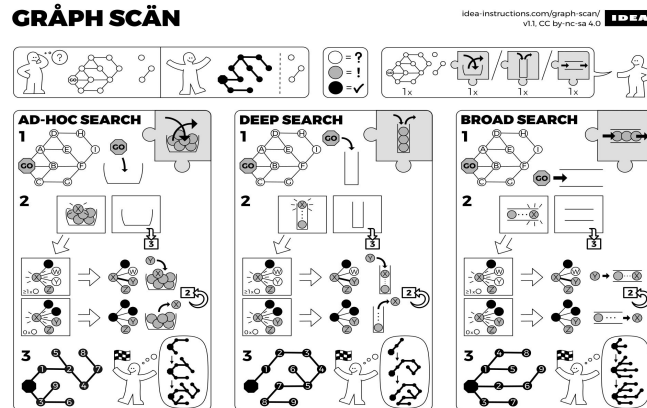


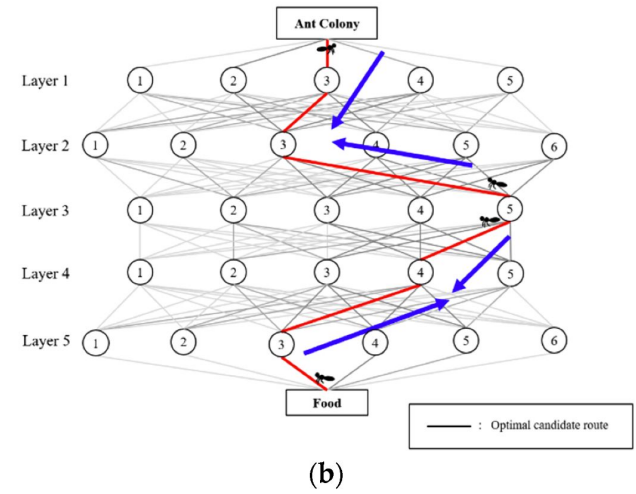
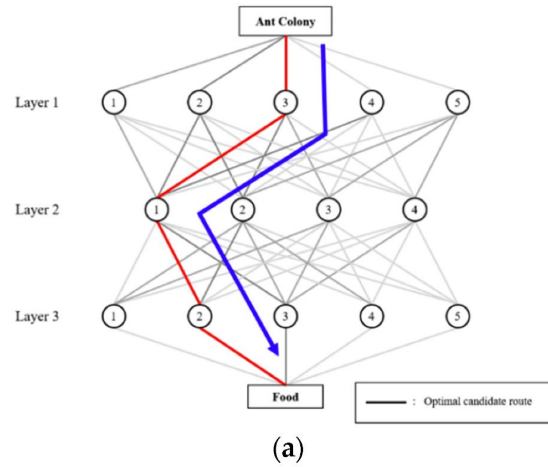
Fig 1.

A: Ants in a pheromone trail between nest and food.

B: an obstacle interrupts the trail.

C: Ants find two paths to go around the obstacle

Network Routing



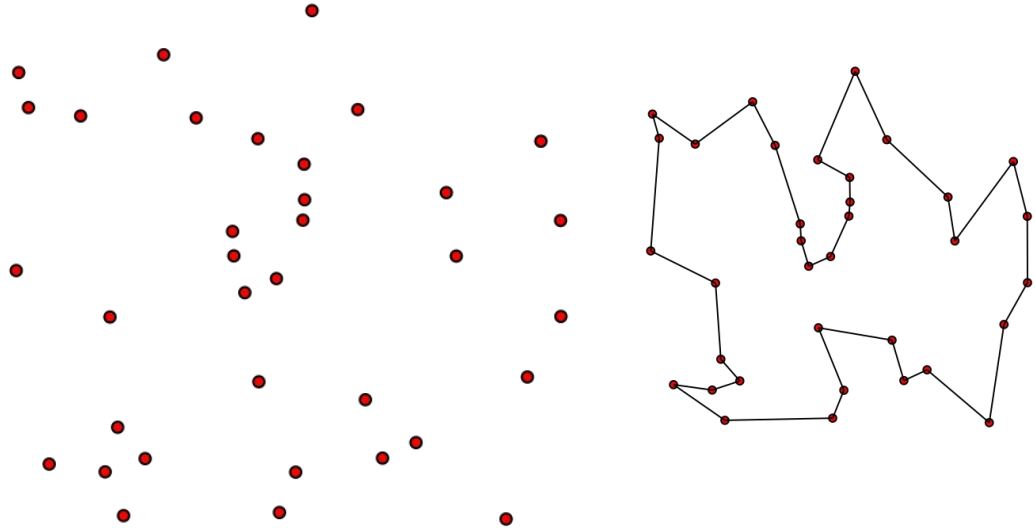
Traveling Salesman problem

"What is the shortest possible route that visits every city once and returns to the starting point?"

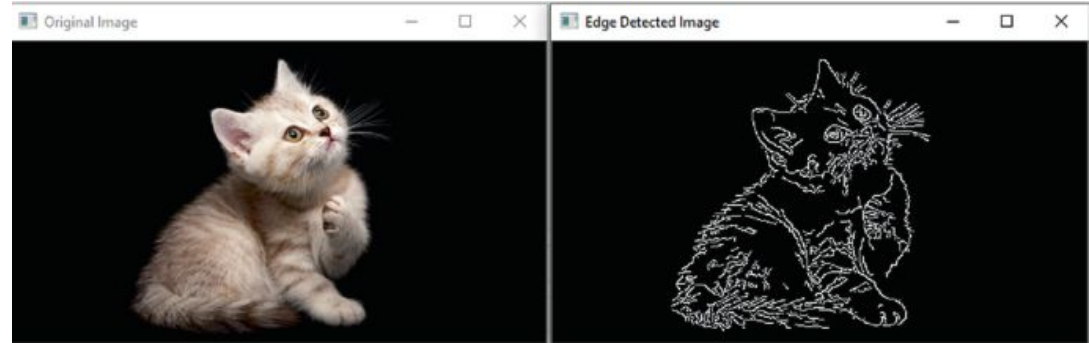
It's extremely hard for computers because the number of routes grows **factorially**.

- 10 cities → 3,628,800 possible routes
- 20 cities → $2.4 * 10^{18}$ possible routes

ACO Approximates a
near-optimal solution rapidly.



EDGE DETECTION IN IMAGES



Unlike traditional algorithms

- ACO can be parralized(GPU)
- Less sensitive the light changes
- Less sensitive to grainy specked backgrounds

ACO_edge_detection.py

Other applications

Job scheduling

Vehicle routing problem (VRP)

Robot path planning

Clustering (data mining)

Feature selection (machine learning)

Image segmentation

Bioinformatics optimization

Power system optimization

Wireless sensor network optimization

Supply chain optimization

Resource allocation



Cool ant videos

Ant bridge

<https://youtu.be/4BdjxYUdJS8?si=wzWj1ZtsYcel-R6c&t=59>

Ant leaf houses

<https://www.youtube.com/watch?v=o3b0U6QOoEg>

Ant raft

https://youtu.be/N_WlgFudDNg?si=htY-LX6aLIY4aeII&t=20

Ant cows

https://youtu.be/xdGwEsXA6P8?si=5LCD_wdzRafY60Q_&t=97